MTH1001

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

MATHEMATICS

May 2020

Mathematical Structures

Module Convener: Professor Nigel Byott

You have 24 hours to complete this paper from the time of its release
Intended duration 2 hours

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is an **OPEN BOOK** examination

SECTION A

1. (a) Using the extended Euclidean Algorithm, calculate $\gcd(129, 17)$ and express it in the form $129a + 17b$ for suitable integers a, b. Hence solve the congruence $17x \equiv 25 \pmod{129}$. (12)

   (b) Let $\sigma = (134)(57)$ and $\tau = (123456)$ in the symmetric group S_7. Find $\sigma\tau$, $\tau\sigma$, τ^2 and $\tau\sigma\tau^{-1}$, writing each as a product of disjoint cycles. Find the order and the sign of each of these permutations. (13)

   (c) What does it mean to say that $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ is a basis in $\mathbb{R}^n$? Show that if $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4$ is a basis of $\mathbb{R}^4$ then $\mathbf{v}_1, \mathbf{v}_1 + \mathbf{v}_2$, $\mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3$, $\mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3 + \mathbf{v}_4$ is also a basis of $\mathbb{R}^4$. (13)

   (d) Consider the following vectors in $\mathbb{R}^4$:

$$\mathbf{w}_1 = \begin{pmatrix} 1 \\ 2 \\ 0 \\ 3 \end{pmatrix}, \quad \mathbf{w}_2 = \begin{pmatrix} 0 \\ 1 \\ -1 \\ 2 \end{pmatrix}, \quad \mathbf{w}_3 = \begin{pmatrix} 1 \\ 0 \\ 2 \\ -1 \end{pmatrix}, \quad \mathbf{w}_4 = \begin{pmatrix} 2 \\ 0 \\ 1 \\ -2 \end{pmatrix},$$

$$\mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}.$$

   Find a maximal linearly independent subset of $\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3, \mathbf{w}_4\}$. Also, determine whether $\mathbf{b}$ can be written as a linear combination of $\mathbf{w}_1$, $\mathbf{w}_2$, $\mathbf{w}_3$, $\mathbf{w}_4$ and, if so, write $\mathbf{b}$ in this way. (12)

[**50**]

SECTION B

2. (a) Let n be a natural number.
   (i) Give the definition of *congruence modulo* n.
   (ii) Show from the definition that if a, b, $c \in \mathbb{Z}$ with $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$ then $a \equiv c \pmod{n}$.
   (iii) Now let m, n be natural numbers, and a, $b \in \mathbb{Z}$. Show that if $a \equiv b \pmod{mn}$ then $a \equiv b \pmod{n}$. (12)

   (b) Let G and H be groups (written multiplicatively). What does it mean to say that a function $\theta : G \to H$ is a homomorphism of groups? (2)

   (c) Determine whether or not each of the following functions is a homomorphism of groups, justifying your answers.
   (i) $\theta : \mathbb{Q}^\times \to \mathbb{Q}^\times$, $\theta(x) = x^2$ for all $x \in \mathbb{Q}^\times$;
   (ii) $\theta : \mathbb{Q}^\times \to \mathbb{Q}^\times$, $\theta(x) = 2x$ for all $x \in \mathbb{Q}^\times$;
   (iii) $\theta : \mathbb{Q}^\times \to \mathbb{Q}^\times$, $\theta(x) = |x|$ for all $x \in \mathbb{Q}^\times$;
   (iv) $\theta : \mathbb{Q}^\times \to \mathbb{Q}^\times$, where

$$\theta(x) = \begin{cases} 1 & \text{if } x > 0, \\ -\frac{1}{2} & \text{if } x < 0. \end{cases}$$

   (Here $\mathbb{Q}^\times$ is the group of *non-zero* rational numbers, under the operation of multiplication.) (11)

   [25]

3. (a) Let V, W be vector spaces over $\mathbb{R}$.
   (i) What does it mean to say that a function $\theta : V \to W$ is a *linear map*?
   (ii) Let U be another vector space over $\mathbb{R}$. Show that if the functions $\phi : U \to V$ and $\theta : V \to W$ are both linear maps, then their composite $\theta \circ \phi : U \to W$ is also a linear map. (11)

   (b) Let G be a group (written multiplicatively).
   (i) Let g, $h \in G$. Show, using mathematical induction, that

$$(hgh^{-1})^n = hg^nh^{-1} \text{ for all } n \in \mathbb{N}.$$

   (ii) Explain what is meant by the *order* $o(g)$ of an element g of G. Your answer should cover both the case where $o(g)$ is finite and the case where $o(g)$ is infinite.
   (iii) Deduce from (i) that if g has finite order $o(g) = m$, then hgh^{-1} also has order m. (14)

   [25]

4. (a) Let $\mathbb{R}^{\mathbb{N}}$ be the vector space over $\mathbb{R}$ consisting of all infinite sequences:

$$\mathbb{R}^{\mathbb{N}} = \{(a_n) : a_n \in \mathbb{R} \text{ for all } n \in \mathbb{N}\}.$$

Determine whether or not the following subsets of $\mathbb{R}^{\mathbb{N}}$ are subspaces. Justify your answers.

(i)
$$V_1 = \{(a_n) \in \mathbb{R}^{\mathbb{N}} : a_n \to 0 \text{ as } n \to \infty\};$$

(ii)
$$V_2 = \{(a_n) \in \mathbb{R}^{\mathbb{N}} : a_n \to \infty \text{ as } n \to \infty\};$$

(iii)
$$V_3 = \{(a_n) \in \mathbb{R}^{\mathbb{N}} : a_{n+1} \leq a_n \text{ for all } n \in \mathbb{N}\};$$

(iv)
$$V_4 = \{(a_n) \in \mathbb{R}^{\mathbb{N}} : a_4 = 0\}.$$

(13)

(b) (i) State carefully the uniqueness property of prime factorisations of natural numbers.

(ii) Using the uniqueness property of prime factorisations, or otherwise, prove that the real number $\sqrt[3]{5}$ is irrational. (12)

[25]